Extra

MA 104

The process of shifting the initial condⁿ to the origin:

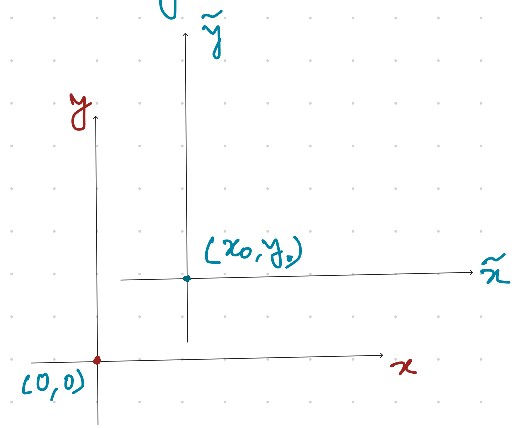
Consider the IVP : $\dfrac{dy}{dx} = f(x,y)$ with $y(x_0) = y_0$. — (1)

Set $\tilde{x} = x - x_0$ & $\tilde{y} = y - y_0$.

Then $$\frac{d\tilde{y}}{d\tilde{x}} = \frac{d\tilde{y}}{dx} \cdot \frac{dx}{d\tilde{x}}$$

$$= \frac{d}{dx}(y - y_0) \cdot \frac{d}{d\tilde{x}}(\tilde{x} + x_0)$$

$$= dy/dx.$$



Hence $\dfrac{d\tilde{y}}{d\tilde{x}} = \dfrac{dy}{dx} = f(x,y) = f(\tilde{x} + x_0, \tilde{y} + y_0)$.

As $x = x_0 \Rightarrow \tilde{x} = x - x_0 = 0$. Similarly $y = y_0 \Rightarrow \tilde{y} = y - y_0 = 0$.

So now the initial condⁿ is : $\tilde{y}(0) = 0$.

New IVP: $\dfrac{d\tilde{y}}{d\tilde{x}} = f(\tilde{x}+x_0, \tilde{y}+y_0)$ with $\tilde{y}(0) = 0$. — (2)

Now $\tilde{y} = \varphi(\tilde{x})$ is a soln of (2) $\Rightarrow$ $y = \varphi(x-x_0)+y_0$ is a soln of (1), since $\dfrac{dy}{dx} = \dfrac{d\,\varphi(x-x_0)}{dx} = \dfrac{d\,\varphi(\tilde{x})}{d\tilde{x}} \cdot \dfrac{d\tilde{x}}{dx}$

$$= f(\tilde{x}+x_0, \tilde{y}+y_0)$$

$$= f(x,y).$$

Eg. Given ODE: $y' = xy^3$ with $y(-1) = 2$. $\rightsquigarrow f(x,y) = xy^3$.

$\Rightarrow$ Set $\tilde{y} = y-2$ & $\tilde{x} = x+1$. Then $\tilde{y}' = f(\tilde{x}-1, \tilde{y}+2) = (\tilde{x}-1)(\tilde{y}+2)^3$.

with $\tilde{y}(0) = 0$.

$$\Rightarrow \frac{d\tilde{y}}{(\tilde{y}+2)^3} = (\tilde{x}-1)\,d\tilde{x} \quad \Rightarrow \quad \frac{1}{-2(\tilde{y}+2)^2} = \tilde{x}^2/2 - \tilde{x} + c.$$

$$\Rightarrow (\tilde{y}+2)^2 = \frac{1}{2\tilde{x} - \tilde{x}^2 - 2c}.$$

Now $\tilde{y}(0)=0 \Rightarrow 4 = -\frac{1}{2c} \Rightarrow c = -1/8$.

Hence $$(\tilde{y}+2)^2 = \frac{1}{2\tilde{x}-\tilde{x}^2-2*(-1/8)} = \frac{1}{2\tilde{x}-\tilde{x}^2+1/4} = \frac{1}{-(\tilde{x}-1)^2+5/4}$$

$$\Rightarrow \tilde{y}+2 = \pm\sqrt{\frac{1}{5/4-(\tilde{x}-1)^2}}.$$

As $\tilde{y}(0)=0$ so $\tilde{y}+2 = \sqrt{\frac{1}{5/4-(\tilde{x}-1)^2}}$, the positive root.

Thus the solⁿ of the given IVP is $y = \frac{1}{\sqrt{5/4-x^2}}$.

Rmk! Similar method can be used for higher order ODEs (with initial condⁿ) as well.

Recall (Uniqueness th^m: $y' = f(x,y)$ with $y(x_0) = y_0$). — (1)

Suppose that f & its partial derivative $\partial f/\partial y$ are cont^n $\forall (x,y)$ in $D: |x - x_0| < a$ & $|y - y_0| < b$ & bounded on D, i.e.,

$$|f(x,y)| \leq K \quad \& \quad |\partial f/\partial y| \leq M \quad \forall (x,y) \in D.$$

Then the IVP (1) has at most one solu^n $\forall x$ with $|x - x_0| < \alpha$, where $\alpha = \min\{a, b/K\}$.

Rmk: Suppose that $f(x,y)$ & $\partial f/\partial y$ are cont^n fu^{n}s $\forall (x,y) \in \mathbb{R}^2$. Now their continuity forces, $f(x,y)$ & $\partial f/\partial y$ both are bounded on $D_L := (x_0 - L, x_0 + L) \times (y_0 - b, y_0 + b)$ for any real no. $L > 0$. For instance we can choose $L = n \in \mathbb{N}_{>0}$. Further assume that $|f(x,y)| \leq c|x|$ for all $(x,y) \in \mathbb{R}^2$ & some const. c.

Set $b = cn$. Then by the uniqueness th^m, $\exists$ a unique sol^n $y_n(x)$ of ① for all x with $|x - x_0| < n$. Now define $y : \mathbb{R} \to \mathbb{R}$

by setting: $y(x) = y_n(x)$ for $x \in (x_0 - n, x_0 + n)$.

Observe that $\bigcup_{n \geq 1} (x_0 - n, x_0 + n) = \mathbb{R}$. The uniqueness of the sol^n in each interval $(x_0 - n, x_0 + n)$ forces

$$y_n(x) = y_m(x) \quad \forall\, m \geq n \text{ on } (x_0 - n, x_0 + n).$$

Thus $y(x)$ is well-defined. The cont^n & diff. of $y(x)$ also follows from the respective properties of $y_n(x)$'s.

Ex. Check $y(x)$ is a sol^n of ①

Ex. Check the def^n of $y(x)$ forces it to be the uniqu. sol^n of ①.